

Compounds of Transitional elements

341. In the metallurgy of zinc, the zinc dust obtained from roasting and reduction of zinc sulphide contains some ZnO. How is this removed
- Absorbance of ultraviolet light and re-emission of white light is employed
 - Shock cooling by contact with a shower of molten lead is done
 - X-ray method is used
 - Smelting is employed
342. In the metallurgy of copper, metallic copper is finally formed in the furnace by the reactions
- $Cu_2S + O_2 \rightarrow 2Cu + SO_2$
 - $2CuS + 3O_2 \rightarrow 2CuO + 2SO_2$
 $2CuO + CuS \rightarrow 3Cu + SO_2$
 - $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$
 $Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$
 - $CuS + O_2 \rightarrow Cu + SO_2$
343. In the smelting of roasted copper pyrites ore, melting occurs so that the first reaction is
- All the sulphur preferentially combines with iron to form FeS and CuO is formed
 - All the sulphur preferentially combines with copper to form CuS and FeO is formed
 - All the sulphur preferentially combines with iron to form FeS and Cu_2O is formed
 - All the sulphur preferentially combines with copper to form Cu_2S and FeO is formed
344. In the oxidation of Cu , the reaction which takes place in bessemer converter is
- $2CuFeS_2 + O_2 \rightarrow Cu_2S + FeS + SO_2$
 - $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$
 - $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$
 - $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$
345. Silica is added to roasted copper ore during smelting in order to remove
- Cuprous sulphide
 - Cuprous oxide
 - Ferrous oxide
 - Ferrous sulphide
346. Parke's process is used to extract
- Silver using $NaCN$
 - Copper using $CuFeS_2$
 - Silver from argentiferous lead
 - Silver by forming amalgam
347. Identify the reaction that doesn't take place during the smelting process of copper extraction
- $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2 \uparrow$
 - $Cu_2O + FeS \rightarrow Cu_2S + FeO$
 - $2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2 \uparrow$
 - $FeO + SiO_2 \rightarrow FeSiO_3$



348. The extraction of which of the following metals involves bessemerisation
 (a) *Fe* (b) *Ag*
 (c) *Al* (d) *Cu*
349. Bessemer converter is used for
 (a) Steel (b) Wrought iron
 (c) Pig iron (d) Cast iron
350. In the cyanide process for the extraction of silver, sodium cyanide is used to
 (a) Convert silver into a soluble silver complex
 (b) Reduce silver
 (c) Precipitate silver
 (d) Oxidise silver
351. Parke's process of desilverization of lead depends upon
 (a) Partition coefficient of silver between molten zinc/molten lead having a high value
 (b) Partition coefficient of silver between molten zinc/molten lead having a low value
 (c) Crystallizing out of pure lead while the silver-lead eutectic which has a lower melting point is left behind in liquid form
 (d) Chemical combination of zinc and silver which precipitates out easily
352. Which is impure form of iron
 (a) Cast iron
 (b) Wrought iron
 (c) Steel iron
 (d) None
353. Blister copper is
 (a) Pure copper
 (b) Ore of copper
 (c) Alloy of copper
 (d) 1% impure copper
354. In the extraction of copper when molten copper is cooled slowly, blister copper is obtained due to evolution of the following gas
 (a) Water vapour
 (b) Sulphur dioxide
 (c) Carbon dioxide
 (d) Carbon monoxide
355. In the electrolytic purification of copper some gold is found in the
 (a) Cathode
 (b) Cathode mud
 (c) Anode mud (d) Electrolyte
356. Purest form of iron is
 (a) Cast iron
 (b) Wrought iron
 (c) Hot steel
 (d) Stainless steel
357. **Spelter** is
 (a) Impure *Cu* (b) Impure *Zn*
 (c) *ZnO* (d) *CuO*
358. A copper coin is completely covered with a gold film and is placed in



- dilute HNO_3 . This will result in formation of
- (a) Gold nitrate
 - (b) Copper nitrate
 - (c) None of these
 - (d) Purple of cassins
359. When zinc is added to $CuSO_4$ copper gets ppt. due to
- (a) Reduction of copper ions
 - (b) Oxidation of copper ions
 - (c) Hydrolysis of copper sulphate
 - (d) Complex formation
360. Addition of iron filings to $CuSO_4$ solution caused precipitation of Cu owing to the
- (a) Reduction of Cu^{++}
 - (b) Oxidation of Cu^{++}
 - (c) Reduction of Fe
 - (d) Reduction of Fe^{+++}
361. Oxygen gas can be prepared from solid $KMnO_4$ by
- (a) Strongly heating the solid
 - (b) Treating the solid with H_2 gas
 - (c) Dissolving the solid in dil. H_2SO_4
 - (d) Dissolving the solid in dil. HCl

